PORTFOLIO RISK REVIEW

Credit Risk Analysis

Identifying high-risk borrower characteristics and evaluating lending strategies to improve portfolio quality

Executive Summary · 255,347 loans · $32.58B portfolio

EXECUTIVE SUMMARY

Where the Portfolio Stands Today

255,347	11.61%	$32.58B	$4.29B
Total Loans	**Default Rate**	**Total Exposure**	**Expected Loss**
Full portfolio, 2024	~1 in 9 borrowers	Full lending volume	13.2% of exposure

Key Findings

- Risk is multidimensional: no single variable strongly predicts default — Age and Interest Rate are the strongest signals, but each explains only a small share of outcomes.
- Income is the clearest lever: low-income borrowers default at 17.4%, nearly double the 9.0% rate of the highest-income segment — while loan sizes are nearly identical across all income levels (~$127.6K).
- A combined loan-sizing and rejection strategy can reduce expected credit losses by roughly $346M (8% of current expected loss) while touching under 5% of total lending volume ($1.51B of $32.58B).

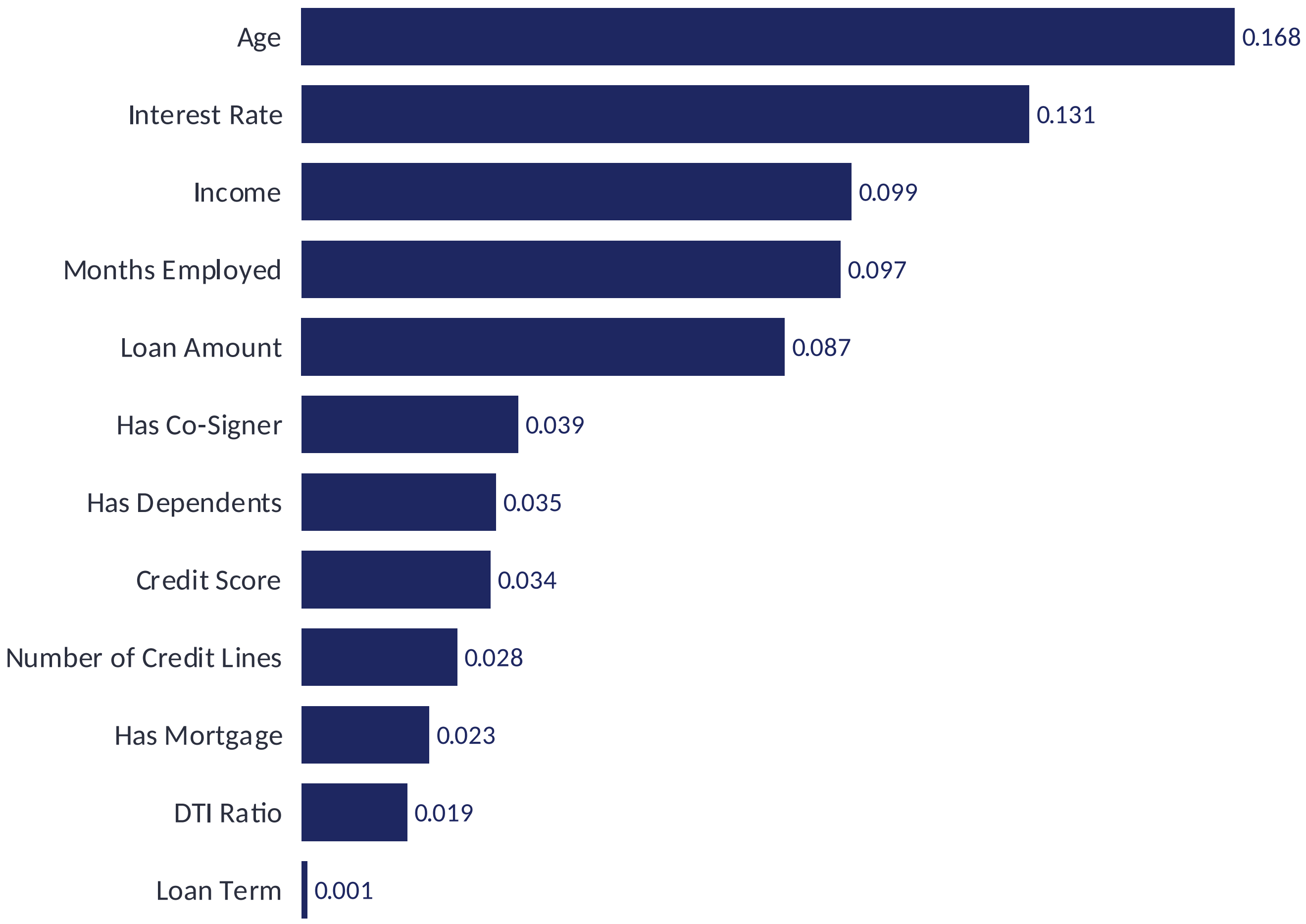
RECOMMENDATION

Hybrid Lending Strategy

- Cap loans 10% for all low-income borrowers
- Cap loans 20% for the ultra-high-risk sub-segment (low income + credit score < 350 + DTI > 0.60)
- Reject the narrow segment meeting all of: credit score < 400, bottom-25% income, DTI > 55%

~$1.51B volume affected • ~$346M expected loss reduction

What Drives Default Risk?



What the Data Tells Us

Age & Interest Rate are the strongest signals — older borrowers and lower rates correlate with lower default risk.

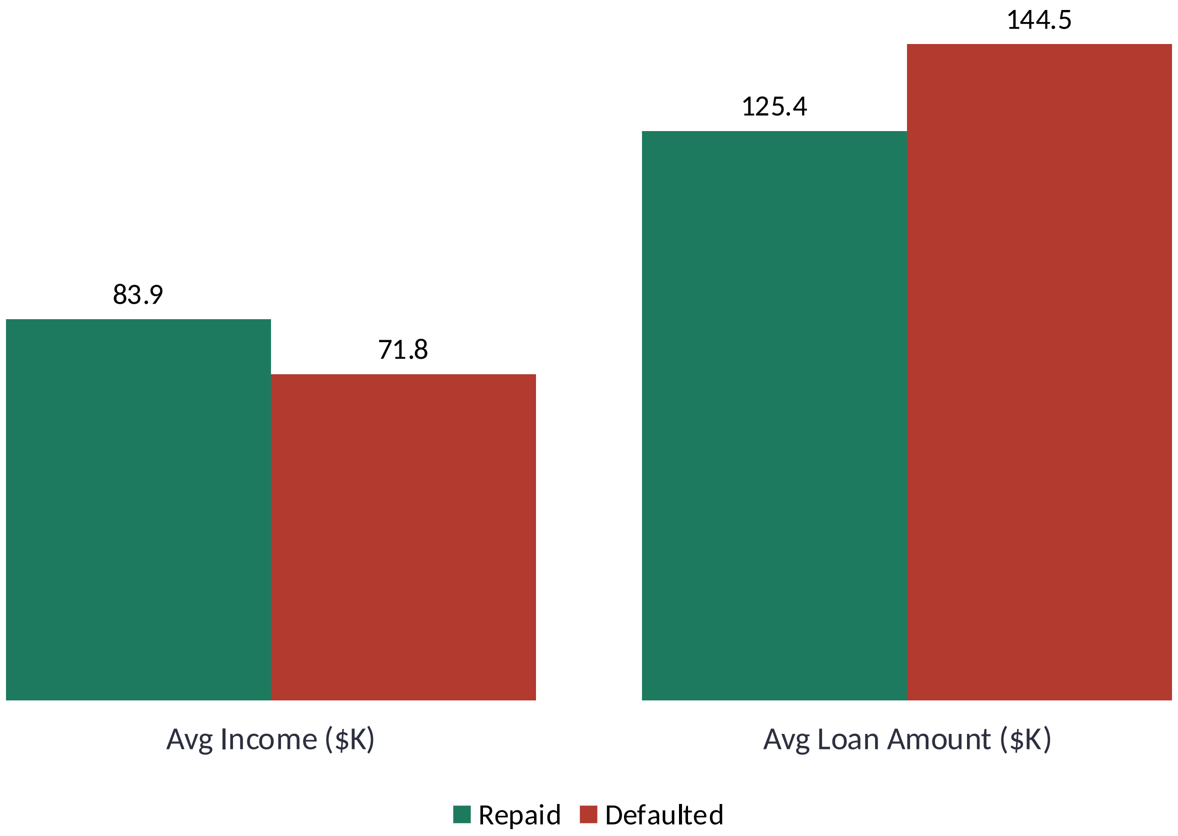
Income & tenure show a moderate protective effect; longer-employed, higher-income borrowers default less.

Credit Score has a weak standalone relationship (0.034) — it ranks 8th of 12 factors and is not a dominant predictor in this portfolio.

No single variable strongly predicts default. Risk is multidimensional — it takes a combination of factors to identify true high-risk borrowers.

Who Are Our Defaulting Borrowers?

Defaulted borrowers share a consistent financial mismatch: lower income, larger loans, higher repayment pressure.



Key Differences

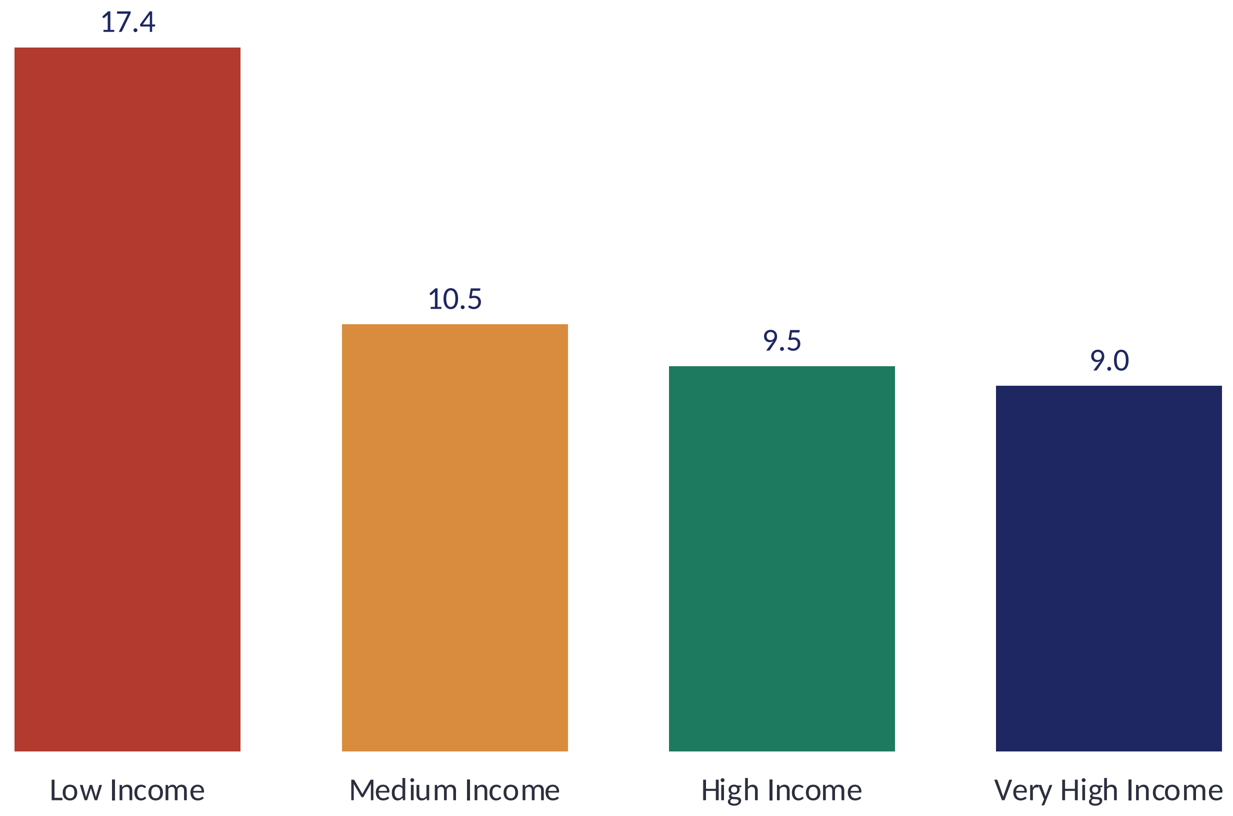
Income gap: Defaulters earn $12.1K less on average ($71.8K vs $83.9K).

Loan size gap: Defaulters borrow $19.2K more on average ($144.5K vs $125.4K).

Credit score: Slightly lower (559 vs 576) — a 17-point gap that alone is not a strong differentiator.

Employment: Unemployed borrowers default at 13.6% vs 9.5% for full-time employees.

Which Borrowers Are Highest Risk?



Inverse Relationship: Income vs. Default

- Low-income borrowers default at 17.4% — nearly 2x the 9.0% rate of the highest-income segment.
- Average loan size is nearly identical across all four income segments (~$127.6K) — loan amounts are not risk-adjusted by income.
- Average credit score is also flat across income segments (~574) — income, not credit score, separates risk here.

Low-income segment: 63,837 borrowers · $8.15B exposure · 17.38% observed default rate.

How Well Does Our Model Perform?

Evaluated on a 51,070-loan holdout test set ($6.53B volume, ~20% of the portfolio) at an approval threshold of 0.20

	Predicted: Decline	Predicted: Approve
Actual: Defaulter	2,624 correctly caught	3,307 missed (risk approved)
Actual: Non-Defaulter	5,905 wrongly declined	39,234 correctly approved

Model correctly flags 44.2% of eventual defaulters (2,624 of 5,931) while approving 83% of all applications — a deliberately volume-friendly setting rather than an aggressive risk filter.

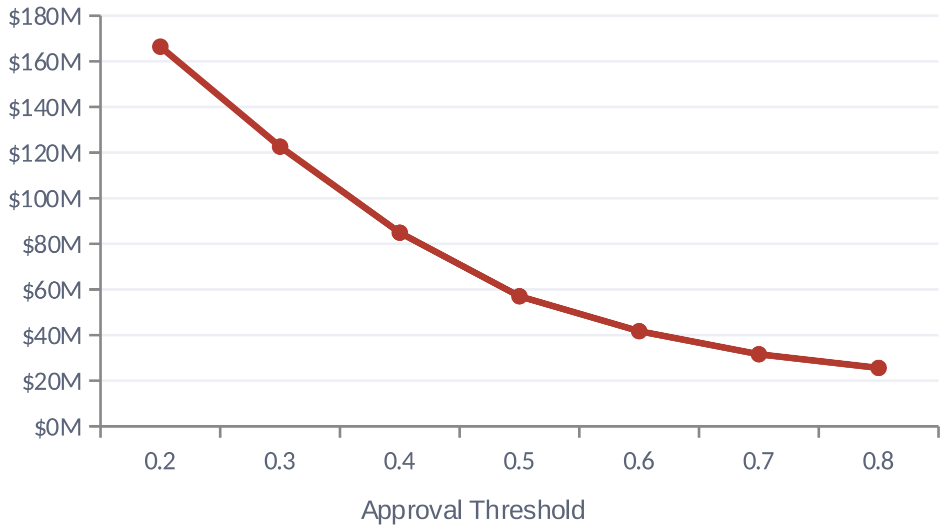
0.751
ROC-AUC

88.6%
Accuracy

83%
Approval Rate

7.8%
Risk in Approved Portfolio (from 11.6%)

Simulated Profit by Approval Threshold



Lower thresholds approve more volume and raise profit but let more risk through; $166.4M at 0.20 is the highest simulated profit tested.

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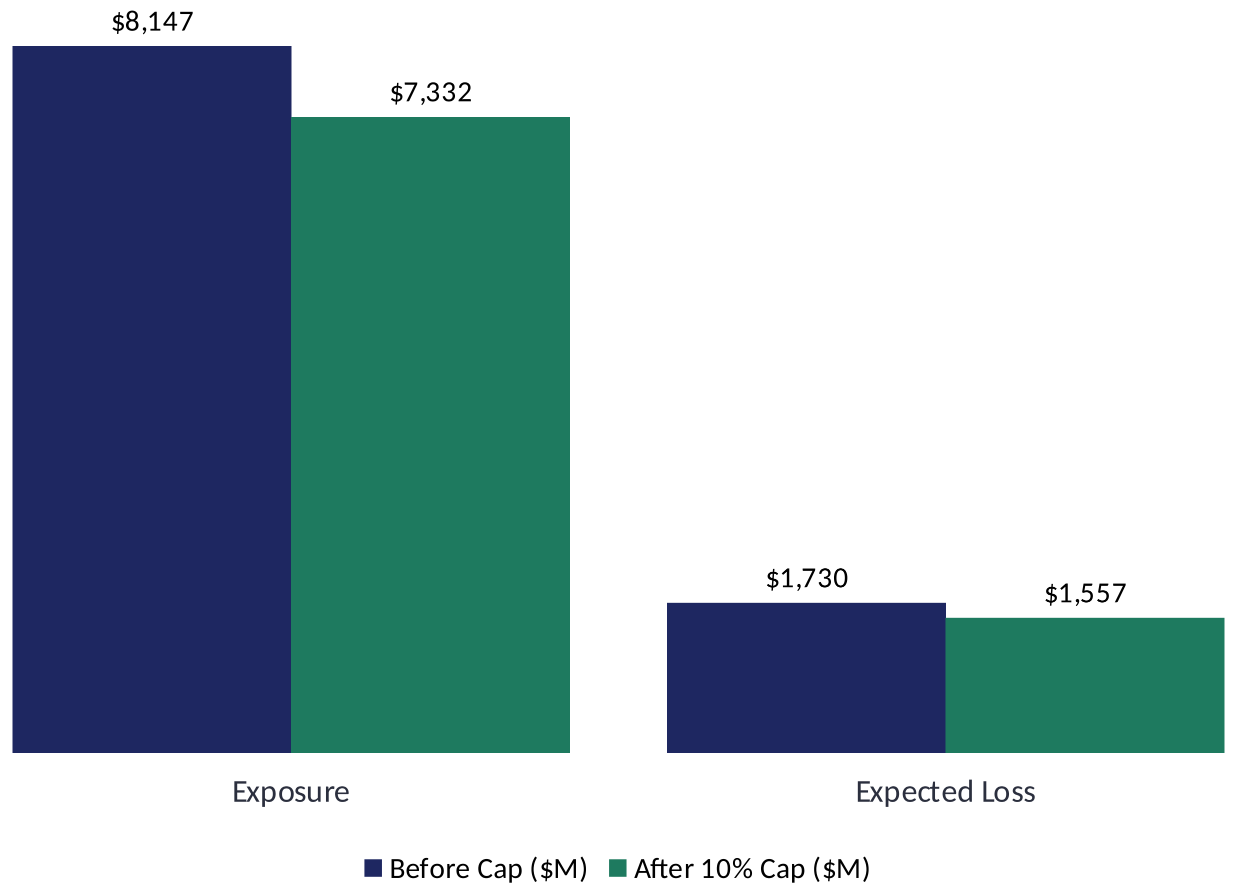
STRATEGY 1

Loan Cap on Low-Income Borrowers



Impact of the Cap

Exposure reduction: $814.7M

Expected loss reduction: $173.2M

Efficiency: 21.3 cents saved per $1 of lending removed

Avg loan per borrower: $127,624 → $114,862

Borrowers affected: 63,837 (all low-income segment)

Loan amounts are uniform across income groups, so capping low-income borrowers directly reduces concentrated risk without restructuring the portfolio.

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STRATEGY 2

Targeted High-Risk Segment Cap

Ultra-targeted filter: **Low Income + Credit Score < 350 + DTI > 0.60** → 2,109 borrowers · $268.3M exposure · 20.6% observed default rate

RECOMMENDED

Scenario A — 10% Cap

Exposure reduced

$26.8M

Expected loss reduced

$6.7M

Efficiency

25.1%

Scenario B — 20% Cap

Exposure reduced

$53.7M

Expected loss reduced

$13.5M

Efficiency

25.1%

Why 20%, Not 10%

- Both scenarios recover the same 25.1 cents of expected loss per dollar of lending removed — efficiency doesn't fall as the cap deepens.
- The deeper 20% cap doubles the loss reduction ($13.5M vs $6.7M) for the same 2,109-borrower segment.
- This is the scenario carried into the final combined recommendation.

Targeting only the highest-risk borrowers produces a higher risk-reduction efficiency than the broad low-income cap alone (25.1% vs 21.3%) — though the total dollar impact is smaller because the segment itself is much smaller.

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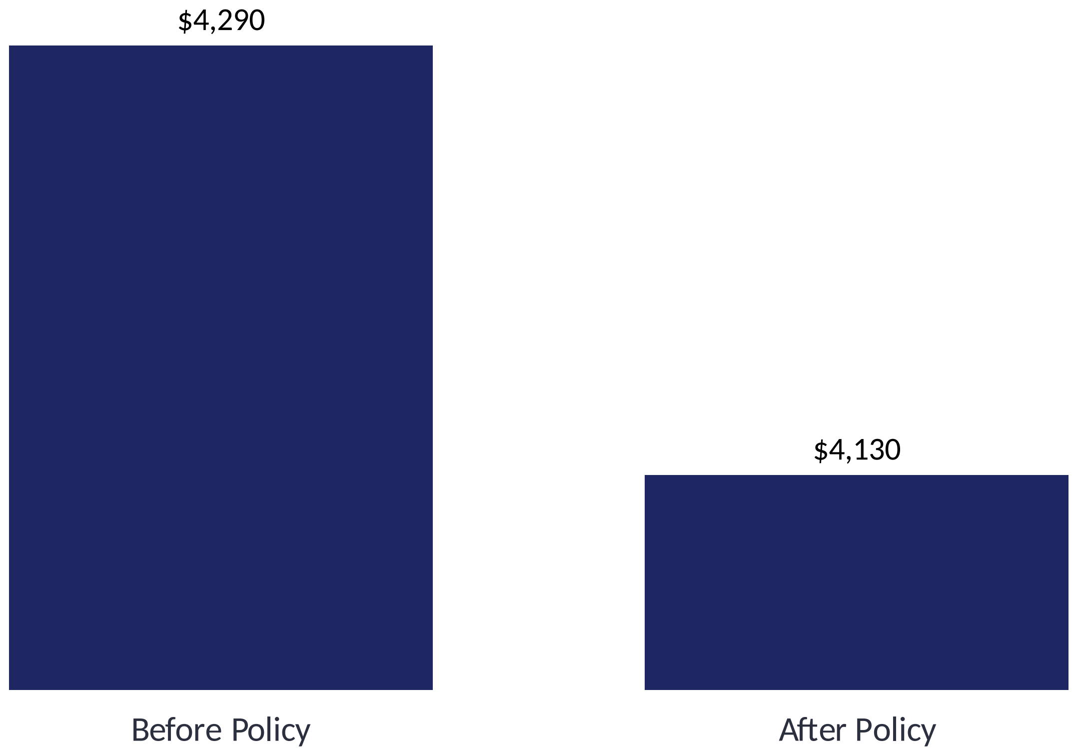
STRATEGY 3

High-Risk Borrower Rejection Policy

Reject applications meeting all three: **Credit Score < 400 • Bottom 25% Income • DTI > 55%**



Portfolio Impact

5,071 applications rejected — a small fraction of the portfolio

$645.6M lending volume removed

$159.7M expected loss avoided (3.7% of total)

11.61% → 11.44% portfolio default rate (-0.17 pp)

A narrow, precisely targeted policy: rejecting the highest-risk 2% of applications avoids a disproportionate share of expected losses.

THE RECOMMENDATION

A Combined, Consistent Hybrid Strategy

	STRATEGY 1 10% cap · all low-income	STRATEGY 2 20% cap · ultra-high-risk	STRATEGY 3 Rejection policy	COMBINED
Criteria	Low income	Low income + Score<350 + DTI>0.60	Score<400 + bottom-25% income + DTI>55%	—
Borrowers affected	63,837	2,109 (subset of Str. 1)	5,071	—
Volume reduced	$814.7M	$53.7M	$645.6M	**$1,514.0M (~$1.51B)**
Expected loss reduced	$173.2M	$13.5M	$159.7M	**$346.4M**
Efficiency (loss $ per $1 removed)	21.3¢	25.1¢	24.7¢	**22.9¢**

Why we can simply add these up: Strategy 2's 2,109 borrowers are a subset of Strategy 1's 63,837 low-income borrowers. The 20% cap replaces Strategy 1's 10% cap for just that subset — it is not stacked on top of it. Under that reading, the three strategies touch non-overlapping slices of exposure and can be summed directly, matching the ~$868M combined exposure figure ($814.7M + $53.7M) cited in the original analysis.

NET RESULT

~$3.94B

Resulting Expected Loss

Down from $4.29B baseline (-8.1%)

11.44%

Portfolio Default Rate

Down from 11.61% — only Strategy 3 changes the default count; Strategies 1-2 cut loss via smaller loan sizes, not fewer defaults